Try

• Give the following prompts to the Generative AI one by one in order:
(1) Give me 5 quiz questions about Egypt (You can ask your school's history teacher for help).
(2) Make them True or False questions.
(3) Display the score after I answer all questions.

• Once you confirm that the quiz has been updated, try changing the quiz topic or answer format, or increasing/decreasing the number of questions.
• Have someone else try the quiz you created.

Also, let’s ask about history like this:
“Who built the pyramids of Egypt?”

Try asking further questions based on the Generative AI’s response.

Check search engines like Google to see if the Generative AI’s output is truly correct (You can ask your school's history teacher for help).

Exercise

Feel free to experiment with generating quizzes or stories, asking study-related questions, etc.

13-2 HTML, CSS

Point!

HTML (HyperText Markup Language) is a language used to display web pages in a web browser. Its role is to create the **structure** of the page, such as headings, tables, and bullet points.

CSS (Cascading Style Sheets) is a language used to define the **design** of a web page. It allows for detailed specification of various designs, such as text size, fonts, and background colors.

By combining HTML and CSS, you can create web pages efficiently.

HTML
Structure and Content
"Headings, text, etc."

Code Example:

```
<h1>My page</h1>
<p>HELLO</p>
<p>WORLD</p>
```

CSS
Design
"Text color, etc."

Code Example:

```
p{
  color : red;
}
```

Browser Display



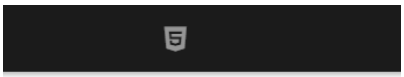
In the example above, the text color inside the HTML <p> tags is set to "Red" using CSS. Because of this, you can change the text color of **all** <p> tags within the HTML from a single location in the CSS.

Warm Up

Let’s write a simple self-introduction page in HTML.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About Me</title>
</head>
<body>
  <h1>Ichiro</h1>
</body>
</html>
```

Copy and paste (or type) the code above into WebDen and run it.



html lang="en"> head> /head>

Adam

/html>

A simple page is displayed.

Next, let’s change the text color and size using CSS.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About Me</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 class="name">Ichiro</h1>
</body>
</html>
```

CSS Code

```
.name {
  color: blue;
}
```

Copy and paste (or type) into WebDen and run it.



The text has turned blue.

In this code, a class named class="name" is specified inside the HTML tag, and the design for that class is defined in the CSS.

By specifying classes in this way, you can individually change the design even for tags of the same type.

Try

Try adding more items (content) to display.
Try changing the text size and font.

Exercise

Let's try creating a page other than a self-introduction page.

13-3 HTML, JS

Point!

JavaScript is a programming language that runs in web browsers.

By embedding it into web pages, you can add interactivity to the page or cause changes to occur in response to user actions.

HTML
Structure and Content
"Headings, text, etc."
Code Example:

```
<h1>My page</h1>
<p id="text1">HELLO</p>
<p id="text2">WORLD</p>
```

JavaScript
Manipulation
"Moving, zooming, changing, etc."
Code Example:

```
document.getElementById("text1").textContent
= "GOOD MORNING";
```

Browser Display



In the example above, an id is specified for the HTML <p> tag, and JavaScript uses that id to change the text inside the tag. **.textContent** is used when specifying (or accessing) the text inside a tag.

Warm Up

Let’s try modifying the self-introduction page using JavaScript.

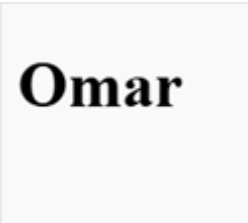
HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About Me</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 id="name">Ichiro</h1>
</body>
</html>
```

JavaScript Code

```
document.getElementById("name").textContent = "Ohtani";
```

Copy and paste (or type) into WebDen and run it.



The name has changed.

Next, let’s try changing the text using a button operation.

HTML Code

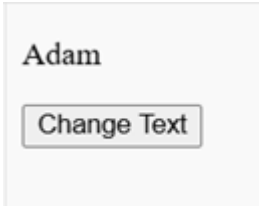
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About Me</title>
</head>
<body>
  <p id="name">Ichiro</p>
  <button id="btn">Change Text</button>
</body>
</html>
```

JavaScript Code

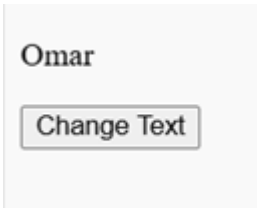
```
const p = document.getElementById("name");
const btn = document.getElementById("btn");

btn.onclick = () => {
  p.textContent = "Ohtani";
};
```

Copy and paste (or type) into WebDen and run it.



When you press the button:



The name changed.

Let’s try changing the text size.

HTML Code

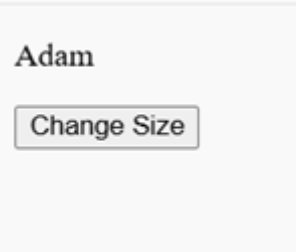
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About Me</title>
</head>
<body>
  <p id="name">Ichiro</p>
  <button id="btn">Change Size</button>
</body>
</html>
```

JavaScript Code

```
const text = document.getElementById("name");
const btn = document.getElementById("btn");

btn.onclick = () => {
  text.style.fontSize = 50 + "px";
};
```

Copy and paste (or type) into WebDen and run it.



When you press the button:



The name’s text became larger.

Try

Try making both the text color and size change when the button is pressed.
Add another item to display, and try creating a function where pressing the button combines (concatenates) the text of the two items.

Exercise

Try making the text color alternate between red and blue each time the button is pressed.

13–4 Creating Static Web Pages

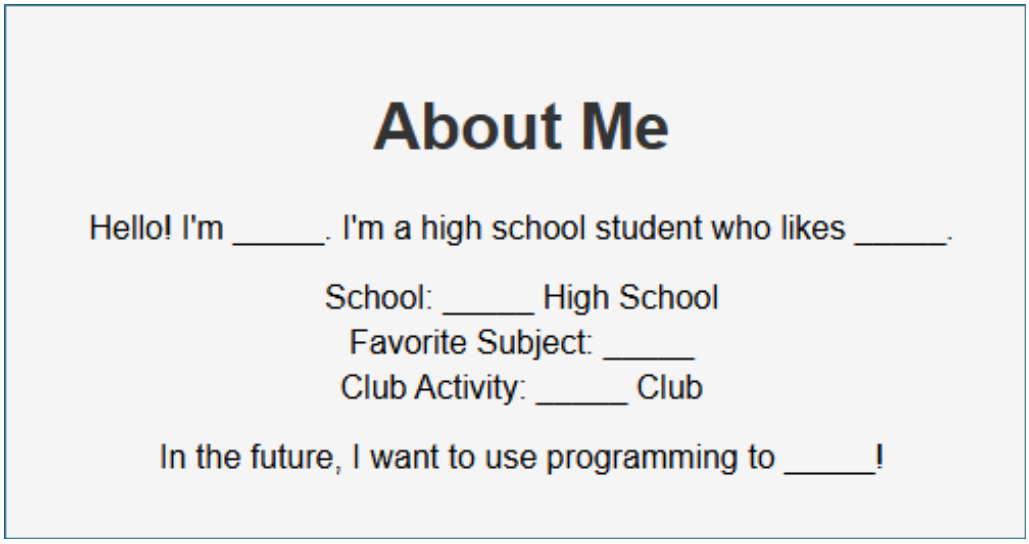
Point!

Programming is one of the strengths of generative AI. You can write programs by giving instructions to the AI. However, just like with text generation, please note that **code generated by AI may contain errors**. Additionally, it may generate code that is beyond your current knowledge, which can be difficult to understand.

Warm Up

Open copilot.microsoft.com in your browser, sign in, and launch Copilot.
1. Generate a Self-Introduction Page Enter the following prompt into Copilot:
“Create a simple self-introduction page using HTML and CSS.”

2. Run the Code Copy and paste the generated output into WebDen and run it.
(Note: Since generative AI produces different code each time, the result is just an example.)

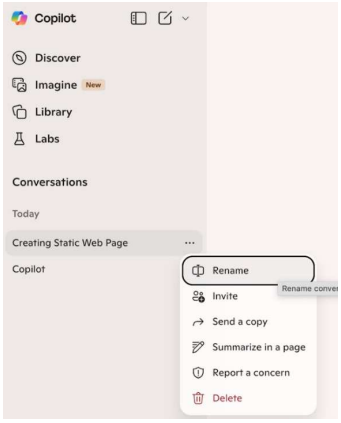


3. Modify the Code Modify the HTML and CSS code to make it your own self-introduction page.

4. Understand the Code Ask the AI to explain the code by using the following prompt:
“Explain this code.”

5. Organize To make it easier to review your interactions in future lessons, rename the Copilot chat to “04 Creating Static Web Pages”.

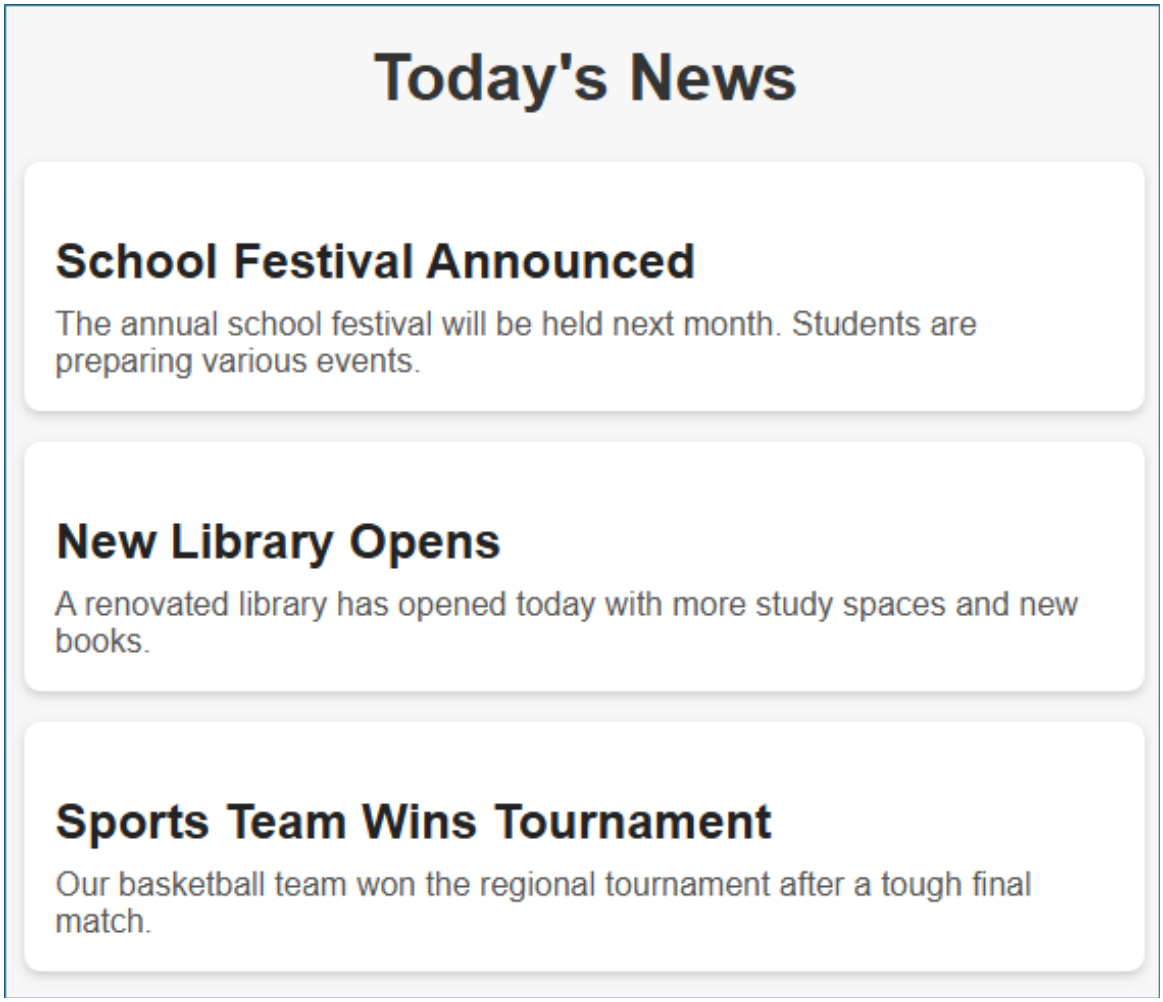
How to rename a chat:
1.Click the “...” (three-dot menu) on the right side of the conversation.
2.Select “Rename” from the menu.
3.Enter the new name and confirm.



Try

Next, try giving the following prompt to the AI:
“Write a simple news page using HTML and CSS.”

1. Run the Code Copy and paste the output into WebDen and run it.
(Note: Results will vary each time.)



2. Customize Modify the code written by the AI. Try changing the text color, font size, background color, etc.

3. Understand Ask the AI for an explanation:
“Explain this code.”

Exercise

Use generative AI to create web pages for various purposes using HTML and CSS.
Examples:

- A ranking of popular sports.
- Recommended restaurants in [Region Name]
- An introduction to [Your School Name].
- Famous tourist spots in Egypt

etc.

13-5 Creating Dynamic Web Pages

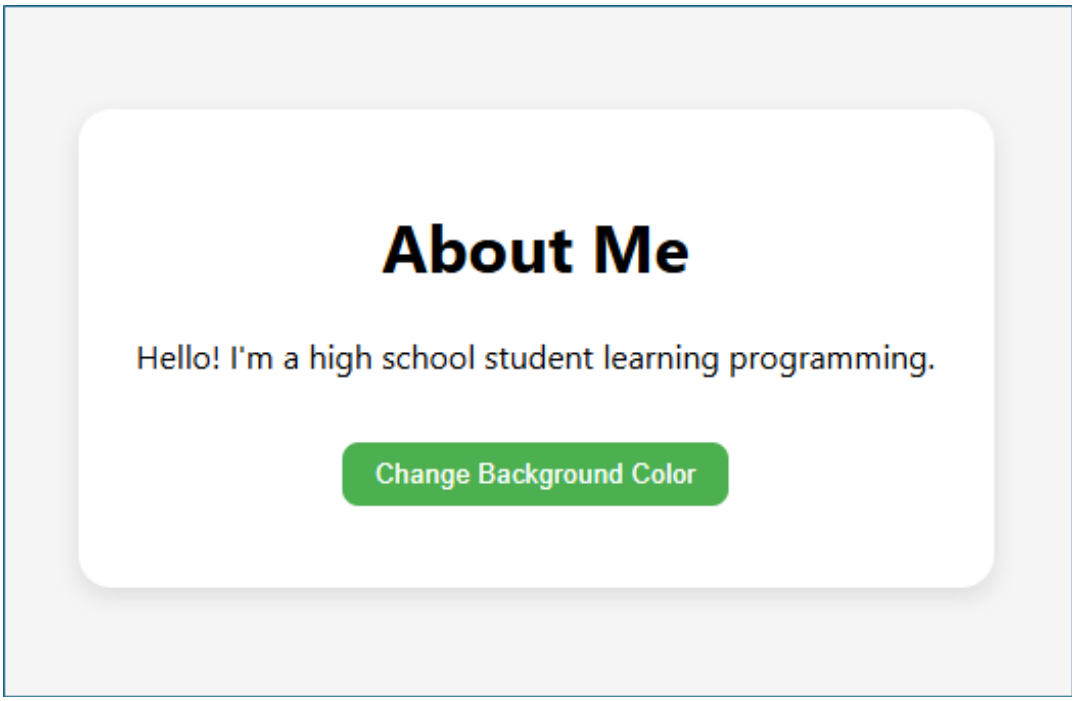
Point!

In this lesson, we will have the Generative AI program **dynamic** (interactive) web pages using JavaScript.

Warm Up

Open “Copilot” and give it the following prompt:
“Create a self-introduction page using HTML, CSS, and JavaScript. Make it so the background color changes when a button is pressed. Please provide separate code for HTML, CSS, and JavaScript.”

Copy and paste the outputted code into WebDen and run it.
(Note: Since using Generative AI results in different code each time, this is just one example.)



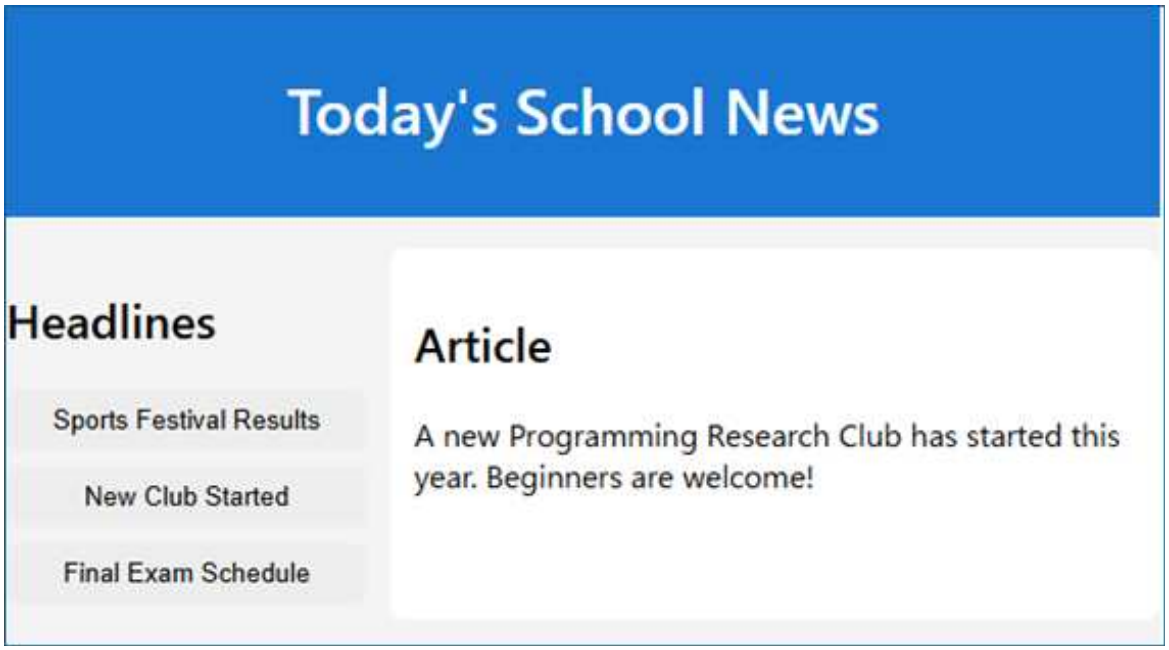
Modify the HTML, CSS, and JavaScript code to change the preset background colors or increase the variety of colors available.

Give the following prompt to the Generative AI and read the explanation: “Explain this code.”

Rename the Copilot chat thread to “Creating Dynamic Web Pages”.

Next, give it the following prompt:
“Write a simple news page using HTML, CSS, and JavaScript. Make the main text switch when a headline is clicked.”

Copy and paste the outputted code into WebDen and run it.
(Note: Since using Generative AI results in different code each time, this is just one example.)



Modify the code written by the Generative AI to add more articles or change the background color depending on the article displayed.

Try

Did you understand the code created by the Generative AI?
Take notes on the parts you don't understand and look them up.
By using Generative AI, you can discover things outside your current programming knowledge.
Let's deepen your programming knowledge by understanding these new concepts.

Exercise

Let’s try creating web pages for various purposes using Generative AI.
Examples:

- Popular sports ranking
- Recommended restaurants in [Region Name]
- Introduction to [School Name] (your school)
- Famous tourist spots in Egypt

etc.

13-6 Presenting Your Website

Point!

In this lesson, you will present the website you created in the previous class to your peers.

Warm Up

Prepare for your presentation by summarizing the key points of your website based on the topics below.
1) What is the theme of your website?
Examples:

• A website about my hobbies.
• A website about soccer.

2) What did you learn about HTML?
Examples:

• CSS and JavaScript can be loaded into HTML using the <link> and <script> tags.
• Assigning an id to a <button> tag allows it to be manipulated by JavaScript.

3) What did you learn about CSS?
Examples:

• You can set the background color using the body selector.
• You can customize button colors and animations using the button selector.

4) What did you learn about JavaScript?
Examples:

• document.getElementById allows you to select specific elements, such as buttons or text, within the HTML.
• addEventListener allows you to define actions to be executed when a button is clicked.

Try

Present your summary to your group members.

13–7 Developing a Quiz App

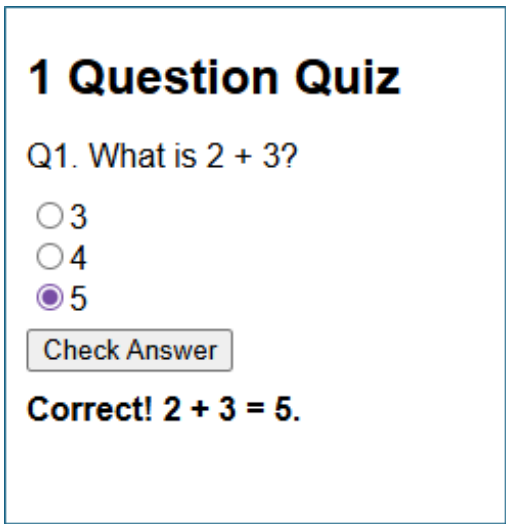
Point!

In this lesson, we will have the Generative AI generate code for a simple Web application using JavaScript.

Warm Up

Open “Copilot” and give it the following prompt:
“Create a simple multiple-choice quiz app. Include only one question. Please provide separate code for HTML, CSS, and JavaScript.”

Paste the outputted code into the HTML, CSS, and JavaScript tabs in WebDen and run it.
(Note: Since using Generative AI results in different code each time, this is just one example.)



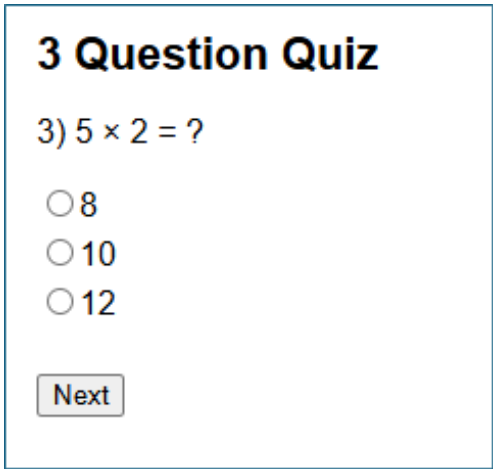
Modify the HTML, CSS, and JavaScript code to change the question text or the answer.

Give the following prompt to the Generative AI and read the explanation:
“Explain this code.”
Rename the Copilot chat thread to “Developing a Quiz App”.

Try

Give the following prompt to the Generative AI:
“Make it 3 questions.”

Paste the outputted code into the HTML, CSS, and JavaScript tabs in WebDen and run it.
(Note: Since using Generative AI results in different code each time, this is just one example.)



Modify the HTML, CSS, and JavaScript code to change the number of questions.

Give the following prompt to the Generative AI and read the explanation:
“Explain this code.”

Exercise

Let’s customize the quiz app you created.
Examples:

- Increase the number of choices (options).
- Display the percentage of correct answers (score).
- Randomize the order of questions.

etc.

13-8 Quiz App Review

Point!

In this lesson, you will summarize what you learned from the quiz app you created in the previous class.

Warm Up

Summarize your key takeaways from the quiz app based on the points below.

1) What is the theme of the quiz you created?
Examples:

• A quiz about space.
• A quiz about animals.

2) What did you learn about HTML?
Examples:

• You can specify the app title using the <title> tag.
• Assigning a class allows you to reference the element in CSS.

3) What did you learn about CSS?
Examples:

• You can set the background color and font using the body selector.
• You can specify the size and color of choice buttons using the .choice class selector.

4) What did you learn about JavaScript?
Examples:

• Multiple options are handled using an array named choices.
• The text color for correct or incorrect answers is changed using an if...else statement.

Try

Categorize the points you summarized into “Things I already knew” and “Things I didn’t know.”
For the things you didn’t know, ask the AI to explain them further to deepen your understanding.

Examples of questions to ask the AI:

• “Tell me what the code does.”

• “What does background: #f5f5f5; mean?”

13–9 Developing a Block Breaker Game 1

Point!

In this lesson, you will use generative AI to generate code for a simple block breaker game using JavaScript.
Note: You will present your work in a future class.

Warm Up

Open copilot.microsoft.com in your browser, sign in, and launch Copilot.

1. Generate the Code Enter the following prompt into Copilot:
   "Create a simple block breaker game. Please provide separate code for HTML, CSS, and JavaScript."

2. Run the Code Copy and paste the generated code into the HTML, CSS, and JavaScript tabs in WebDen and run it.
   (Note: Since generative AI produces different code each time, your result may differ from others.)



3. Understand the Code Give the following prompt to the AI and read its explanation:
   "Explain this code."

How much of the AI-written code can you understand?

Note: Rename the Copilot chat to "Block Breaker Development" to make it easier to review your interactions in future lessons.

Try

Try making the following changes to the game. For this task, do not use generative AI. Try to figure out the logic and modify the program manually by yourself.

• Make the ball larger.

• Change the color of the blocks.

• Increase the number of blocks.

etc.

Exercise

Reflect on your experience modifying the program and write down what you learned.
1) What was difficult?
Examples:

• I didn’t know which value to change to make the ball bigger.
• It was hard to find where the ball was being drawn in the code.

2) What strategies did you use?
Examples:

• I guessed the meaning from variable names to find relevant parts.
• I changed values little by little and adjusted them while checking the movement.

13-10 Developing a Block Breaker Game 2

Point!

In this lesson, you will use AI to update the block breaker game you created last time.
Note: You will present your work in a future class.

Warm Up

Open Copilot and access the “Block Breaker Development” chat you created in the previous lesson.

1. Update the Code Run a prompt like the following:
   “Make the ball change its angle every time it hits the player.”

2. Run the Code Copy and paste the generated code into the HTML, CSS, and JavaScript tabs in WebDen and run it.
   (Note: Since generative AI produces different code each time, this is just an example.)



Try

Try making further changes to the game, such as:
• Changing the ball speed.
• Displaying a score.
• Setting a number of lives.
• Changing the durability of the blocks (making them harder to break).
etc.

You may use AI to generate the code or modify it yourself manually.

Exercise

Prepare for your next presentation by summarizing your work based on the following points:
• What kind of game did you create?
• What were the creative points (strategies you used) and what difficulties did you face?
• What can AI do, and what can it not do?

13-11 Block Breaker Summary

Point!

In this lesson, you will prepare to present the block breaker game you created in the previous class.
Note: You will present your work in the next class.

Warm Up

Reflect on the block breaker app you created in the last lesson and write down your thoughts on the following points.

1) What kind of game did you create?
• Game Title
• Game Rules
Examples:

Number of lives, rankings, presence of items, etc.

2) Development Stories
• Creative Points (Strategies used)
Examples:

I made the ball speed change every time it hits the player.
I made the score text color change when a high score is reached.

• Difficulties
Examples:

Increasing the number of blocks made the game too difficult.
The program stopped working halfway through, so I had to start over from scratch.

3) Collaboration with AI
• What AI can do.
• What AI cannot do.

You may use AI to help you think about the points above.

Try

Organize what you wrote down into a presentation format.
Summarize it into 4 to 5 slides.
You may use AI to help you summarize.

Exercise

Practice your presentation for the next class.

13-12 Block Breaker Development Presentation

Point!

In this lesson, you will present the block breaker game you developed in the previous classes.

Warm Up

You will present the block breaker game created in previous lessons based on the following points.

1) What kind of game did you create?
• Game Title
• Game Rules

2) Development Stories
• Creative points (Strategies used)
• Difficulties faced

3) Collaboration with AI
• What AI can do
• What AI cannot do

Try

Present your summary to your group members.

EDUCATION SPRINGS NEW LIFE



